

API Integration – Weather Information

InternSpark Internship Task 2

1. Project Objective

The objective of this project is to integrate a public weather API with Python and retrieve current weather information for a city entered by the user.

2. Technologies and API Used

Python 3 and the requests library were used. The project uses the Open-Meteo Geocoding API to find the city's coordinates and the Open-Meteo Forecast API to retrieve current temperature, humidity and wind speed. No API key is required.

3. How the Project Works

- The user enters a city name.
- The geocoding API converts the city name into latitude and longitude.
- The weather API is called using those coordinates.
- The current temperature, humidity and wind speed are read from the JSON response.
- The weather report is displayed in the Colab output.

4. Source Code

```
import requests

# Ask the user for a city
city = input("Enter city name: ").strip()

# Find the city's latitude and longitude
geo_url = "https://geocoding-api.open-meteo.com/v1/search"

geo_params = {
    "name": city,
    "count": 1,
    "language": "en",
    "format": "json"
}

geo_response = requests.get(geo_url, params=geo_params)
geo_data = geo_response.json()

if "results" not in geo_data:
    print("City not found. Please try again.")
else:
    place = geo_data["results"][0]

    latitude = place["latitude"]
    longitude = place["longitude"]
    city_name = place["name"]

    # Get current weather
    weather_url = "https://api.open-meteo.com/v1/forecast"

    weather_params = {
        "latitude": latitude,
        "longitude": longitude,
        "current": "temperature_2m,relative_humidity_2m,wind_speed_10m",
        "timezone": "auto"
    }

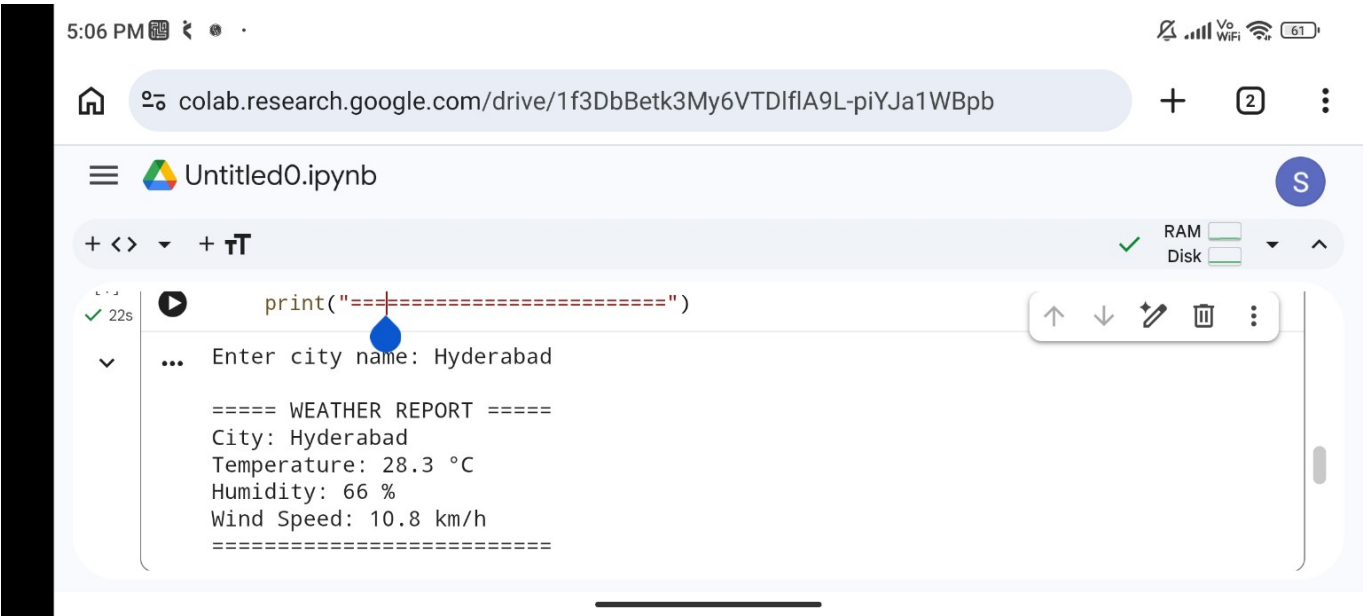
    weather_response = requests.get(
        weather_url,
        params=weather_params
    )
    weather_data = weather_response.json()
```

```
current = weather_data["current"]

print("\n==== WEATHER REPORT =====")
print("City:", city_name)
print("Temperature:", current["temperature_2m"], "°C")
print("Humidity:", current["relative_humidity_2m"], "%")
print("Wind Speed:", current["wind_speed_10m"], "km/h")
print("=====")
```

5. Execution Evidence

The following is the actual Google Colab execution screenshot for Hyderabad. It shows the weather values returned by the API.

A screenshot of a Google Colab notebook titled 'Untitled0.ipynb'. The interface shows a code cell with the command `print("=====")` and its output. The output consists of a prompt 'Enter city name: Hyderabad', followed by a separator line '==== WEATHER REPORT ====', and then the weather details: 'City: Hyderabad', 'Temperature: 28.3 °C', 'Humidity: 66 %', and 'Wind Speed: 10.8 km/h'. The output is enclosed in another separator line '====='. The Colab interface includes a top bar with the URL 'colab.research.google.com/drive/1f3DbBetk3My6VTDIfIA9L-piYJa1WBpb', a file explorer on the left, and a status bar on the right showing RAM and Disk usage. The execution time for the cell is 22 seconds.

```
print("=====")
```

✓ 22s

... Enter city name: Hyderabad

==== WEATHER REPORT ====

City: Hyderabad

Temperature: 28.3 °C

Humidity: 66 %

Wind Speed: 10.8 km/h

=====

6. Result

The application successfully connected to the weather API and displayed the current weather report for Hyderabad, including temperature, humidity and wind speed.

7. GitHub / Public Project Link

Add the real public GitHub repository link here after uploading the Task 2 code.

GitHub Repository: _____

8. Conclusion

This project demonstrates practical API integration using Python, HTTP GET requests and JSON data handling. It provides real-time weather information based on the city entered by the user.